

AyushCase — Technology Stack & Implementation Summary

Smart India Hackathon 2026 | Ministry of Ayush Problem Statement

1. Complete Technology Stack in Brief

- Frontend Framework:** **Next.js 14 (App Router), React 18**
Server-side rendering (SSR) and dynamic client components with seamless routing, React hooks, and context state.
- Styling & Design System:** **Tailwind CSS, Glassmorphism, Vanilla CSS Tokens**
Modern clinical responsive layout, dark/light emerald accents, accessible UI badges, and animated feedback states.
- Database & ORM:** **Prisma ORM 5.21, SQLite (PostgreSQL Ready)**
Declarative schema management with relations for Doctor, Patient, CaseRecord, and MedicalDocument models.
- Authentication & Security:** **NextAuth.js v4 (JWT Strategy), Bcrypt.js**
Secure session cookies, encrypted password hashing, and role-based doctor/patient access control.
- AI & LLM Architecture:** **Hybrid Multi-Model (Gemini 1.5 Flash, GPT-4o, Classical Fallback)**
Dynamic model orchestration with strict domain guardrails, structured JSON output parsers, and zero-latency offline knowledge fallback.
- Voice & Speech Engine:** **Native Browser Web Speech API (webkitSpeechRecognition)**
Real-time bilingual voice dictation (Hindi & English), on-device speech processing, and zero audio API cost.
- OCR & Document Extraction:** **Custom Clinical Named-Entity Recognition (NER)**
Multi-entity parser for prescriptions, lab tests, reference range comparisons, and HIGH/LOW abnormality flags.
- PDF Generation & Export:** **jsPDF 2.5, html2canvas**
Automated client/server PDF generation for printable Ayurvedic clinical case sheets, medical passports, and reports.
- Deployment & Cloud:** **Vercel Serverless Platform, Git CI/CD**
Continuous deployment with automated prisma client generation and self-healing serverless database auto-seeding.

2. Key Functional Modules Implemented

&j Patient Portal & AI Pre-Consultation Intake

- Bilingual interface (English & Hindi) with explicit digital informed consent.
- 1-Click Demo Auto-fill (Rajesh Kumar) for instant presentation flow.
- Captures demographics, blood group, contact, and official 14-digit ABHA ID.

&j Conversational AI History-Taking & Emergency Triage

- Voice & typed conversation progressively covering 7 clinical stages (Complaint, HPI, Past, Meds, Family, Lifestyle, Agni).
- Strict scope constraint: Collects history only without issuing autonomous diagnoses.
- Synchronous Red-Flag Emergency detection for cardiac, stroke, respiratory, and acute hemorrhage symptoms.

&j OCR Document Digitization & Medical Timeline

- Uploads prescriptions, lab panels, and discharge summaries with 1-click sample presets.
- Extracts medicines, dosages, and lab parameters with abnormal threshold indicators.
- Chronological Medical Timeline interleaving historical visits and external OCR records.

&j Physician-Ready Summary & 1-Click Doctor Pre-fill

- Synthesizes conversational history + OCR records with mandatory "Doctor verification required" disclaimer.
- 1-Click "&j Start Consultation" pre-populates case sheet, saving ~15 minutes of doctor docum

&j Ayush AI Floating Assistant (4 Specialized Personas)

- AyurVaidya (Classical Ayurveda & Doshas), Clinical Pariksha (Doctor Guide), AyushGuard (Herb-Drug Safety), AyushCare (Patient Companion).
- Strict Domain Boundary: Filters out non-medical and non-AYUSH questions with polite redirection.
- Strict Project Identity: Identifies exclusively as the AyushCase Clinical AI Assistant.

&j AYUSH Clinical Intelligence & ABDM Interoperability

- 8-Parameter Prakriti algorithm with dynamic Vata-Pitta-Kapha scoring.
- Ashtavidha & Dashavidha Pariksha structured physical examination matrices (Nadi, Jihva, Mala, Mutra).
- Dual Diagnostic Mapping linking classical Sanskrit terms with WHO ICD-11 codes for ABDM interoperability.